KONGU ENGINEERING COLLEGE

TRANSFORM YOURSELF

EV-Optimized Route Planning Platform

Software Requirements Specification

Document Version: **1.0**

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Abstract

The EV-Optimized Route Planning Platform is an AI-powered navigation system designed to address the critical challenge of range anxiety faced by electric vehicle (EV) drivers. By integrating real-time data on traffic conditions, terrain elevation, weather patterns, and charging infrastructure availability, the platform generates routes that minimize battery consumption while ensuring drivers reach their destinations reliably. The system serves individual EV drivers, fleet operators, and charging network stakeholders who require dependable, energy-efficient routing decisions. As global adoption of electric vehicles accelerates, tools that reduce the practical barriers to EV ownership become increasingly important. This platform directly addresses that need by combining predictive energy modeling with dynamic charging stop recommendations, thereby improving driver confidence, reducing unnecessary energy expenditure, and supporting the broader transition toward sustainable transportation.

Background

Electric vehicle adoption has accelerated substantially over the past decade, driven by falling battery costs, expanding model availability, and growing regulatory pressure to reduce transportation-related carbon emissions. Despite this momentum, range anxiety — the fear that a vehicle's battery will be depleted before reaching a destination or a suitable charging point — remains one of the most persistent barriers to widespread EV adoption. Surveys of prospective and current EV owners consistently identify uncertainty about driving range and charging availability as a leading concern, one that often outweighs considerations such as purchase price or vehicle performance.

Drivers face several concrete pain points in their day-to-day use of electric vehicles. Route planning tools rarely account for the factors that most affect EV energy consumption, leaving drivers to guess whether a given route is achievable on the remaining charge. Visibility into charging station location, availability, and compatibility is fragmented across multiple apps and websites, forcing drivers to cross-reference several sources before committing to a route. Furthermore, energy usage is highly sensitive to variables that conventional navigation systems

ignore entirely, including road gradient, ambient temperature, wind conditions, and stop-and-go traffic, all of which can cause actual range to deviate significantly from the vehicle's rated range.

Existing navigation systems were designed primarily for internal combustion engine vehicles, where refueling is fast and widely available, and where energy efficiency has a comparatively small effect on trip feasibility. These systems typically optimize for shortest time or shortest distance without regard to battery state of charge, terrain-induced energy draw, or the strategic placement of charging stops. As a result, EV drivers using conventional tools risk being routed along paths that are technically shorter but practically infeasible given their vehicle's remaining energy. This gap between what generic navigation tools offer and what EV drivers actually need represents a significant and growing problem as EV fleets expand. Addressing it also supports the broader global push toward sustainable transportation, since more reliable, range-aware navigation removes a key psychological and practical obstacle to shifting drivers away from fossil-fuel vehicles.

Problem Statement

Electric vehicle drivers lack a navigation tool that reliably minimizes battery consumption while accounting for the full range of factors that influence real-world energy usage. Route planning for EVs is a multi-factor optimization problem that must simultaneously weigh traffic congestion, road terrain and elevation change, weather conditions, the availability and compatibility of charging infrastructure, and the specific energy characteristics of the vehicle being driven. Each of these factors interacts with the others in ways that are difficult to predict manually, and their combined effect on energy consumption can vary substantially from one trip to the next.

This is a genuinely hard problem because it depends on real-time and highly variable data streams — live traffic, live weather, and live charging station status — that must be synthesized into a single optimized route within a short response window. No unified system today combines all of these inputs into a single, coherent routing decision; drivers are instead forced to reconcile information from disparate sources themselves, a process that is error-prone and time-consuming. The proposed EV-Optimized Route Planning Platform will solve this problem by using an AI-driven optimization engine that ingests all relevant data sources and produces a single battery-

efficient route with proactively recommended charging stops, removing the guesswork from EV trip planning.

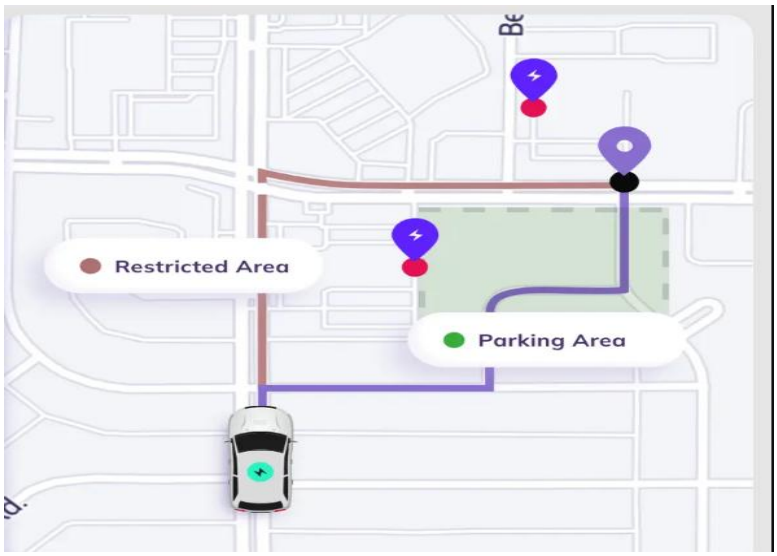


Figure 1. The EV, driver, and environmental factors (traffic, terrain, weather) feeding into the route optimization problem alongside charging infrastructure availability.

Impact

- **Reduce EV range anxiety:** By providing drivers with a route that has already accounted for battery consumption and charging needs, the platform removes the uncertainty that causes range anxiety. Drivers gain confidence that the suggested route is achievable, which directly supports greater willingness to drive EVs on longer or unfamiliar trips.
- **Improve battery utilization:** The routing engine actively selects paths and driving conditions that minimize unnecessary energy draw, such as avoiding steep grades or heavy congestion where alternatives exist. This translates into fewer charging stops per trip and lower overall energy cost for the driver.
- **Increase EV adoption:** As the platform makes EV ownership more practical and less stressful for day-to-day and long-distance driving, it removes one of the primary psychological barriers that currently deters prospective buyers, thereby supporting continued growth in the EV market.

- **Support sustainable transportation:** By making electric vehicles more usable and dependable, the platform contributes to broader environmental and policy goals around reducing transportation emissions, aligning directly with regional and global sustainability targets.

Key Deliverables

1. **Generate Battery-Optimized Routes:** The platform will calculate driving routes that minimize total battery consumption rather than simply minimizing distance or travel time. The routing engine will evaluate multiple candidate paths and select the one offering the best balance of energy efficiency and trip duration, giving drivers a dependable, ready-to-use route for every journey.
2. **Predict Energy Consumption Accurately:** Using a predictive model trained on vehicle characteristics, terrain, and driving conditions, the system will estimate how much battery charge a given route will consume before the trip begins. This allows drivers to verify feasibility and plan charging stops proactively rather than reactively.
3. **Recommend Charging Stops Dynamically:** When a route requires recharging, the platform will identify and suggest charging stations along the way based on real-time availability, connector compatibility, and minimal detour time, ensuring the recommendation fits naturally into the planned trip.
4. **Consider Weather and Terrain Factors:** The platform will incorporate live weather data and elevation profiles into its energy model, since factors such as headwinds, cold temperatures, and steep climbs materially affect battery drain. This ensures route recommendations remain accurate under real driving conditions rather than idealized ones.
5. **Provide Real-Time Route Updates:** As conditions change during a trip — new traffic incidents, station outages, or weather shifts — the platform will recalculate and update the route in near real time, keeping the driver on the most energy-efficient path available at every stage of the journey.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete set of functional and non-functional requirements for the EV-Optimized Route Planning Platform. It describes the intended behavior of the system, the interfaces it exposes to users and external services, and the constraints under which it must operate. This document is intended to serve as the authoritative reference throughout the design, development, testing, and evaluation phases of the project, ensuring that all stakeholders share a common understanding of the system's scope and expected functionality.

1.2 Document Conventions

Throughout this document, requirement identifiers are presented in bold (for example, REQ-1.1) to allow unambiguous traceability. Italicized text is used to denote explanatory notes, assumptions, or clarifying remarks that are not themselves requirements. The word "shall" is used exclusively to denote a mandatory requirement that the system must satisfy; the words "should" and "may" are used to denote recommended or optional behavior, respectively. Figure and table placeholders follow a consistent numbering scheme that is sequential across the entire document.

1.3 Intended Audience and Reading Suggestions

This document is intended for a range of stakeholders, including software developers responsible for implementing the system, project mentors and academic evaluators assessing the project's completeness and technical soundness, and external or peer reviewers who require a clear understanding of the platform's intended functionality. Developers should focus primarily on Sections 3 through 5, which contain the detailed interface and functional requirements. Evaluators and reviewers are encouraged to begin with Sections 1 and 2 for context before proceeding to the detailed requirements in later sections. Readers seeking a high-level overview of the project's goals

and rationale should also refer to the Background, Problem Statement, and Impact sections that precede this Introduction.

1.4 Product Scope

The EV-Optimized Route Planning Platform is an AI-driven navigation application that generates battery-efficient routes for electric vehicle drivers. The platform accepts a desired origin and destination, evaluates candidate routes against real-time traffic, terrain, and weather data, predicts energy consumption along each candidate, and recommends the route and charging stops that best balance energy efficiency and trip duration. The system's scope extends to individual EV drivers planning personal trips, though its underlying architecture is designed with future extensibility toward fleet-operator use cases in mind. By reducing the uncertainty associated with EV trip planning, the platform directly supports the broader goal of encouraging sustainable transportation and lowering barriers to electric vehicle adoption.

1.5 References

Reference Title	Author / Source	Version	Date
IEEE Recommended Practice for Software Requirements Specifications (IEEE Std 830-1998)	IEEE Computer Society	830-1998	1998
Schema.org Vocabulary for Structured Web Data	Schema.org Community Group	Latest	Ongoing
OpenChargeMap API Documentation	OpenChargeMap.org	Latest	Ongoing
National EV Charging Infrastructure Guidelines	[Insert Governing Body]	TBD	TBD
Web Content Accessibility Guidelines (WCAG)	W3C	2.1	2018

2. Overall Description

2.1 Product Perspective

The EV-Optimized Route Planning Platform is a new, standalone AI-driven navigation system. While it does not depend on any pre-existing internal system, it is designed to integrate with established external ecosystems in order to provide accurate, real-time routing decisions. Specifically, the platform draws on third-party mapping services for base road-network data, weather data providers for live meteorological conditions, charging network APIs for station location and availability, and vehicle telematics interfaces for real-time battery state of charge. The platform's optimization engine sits at the center of this ecosystem, consuming data from each external source and producing a single, unified route recommendation for the driver.

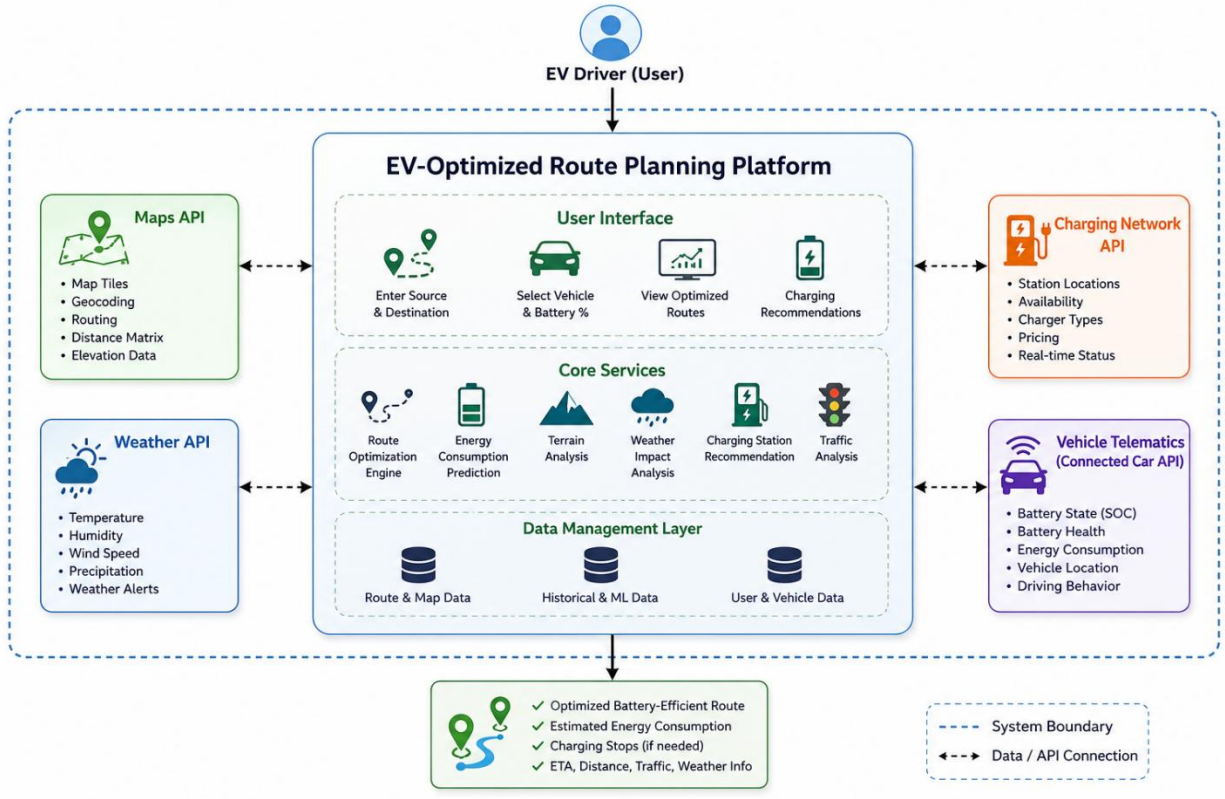


Figure 2. The EV-Optimized Route Planning Platform and its boundaries, showing external systems it connects to: maps API, weather API, charging network API, and vehicle telematics.

2.2 Product Functions

- Generate battery-optimized driving routes between a specified origin and destination.
- Predict energy consumption for candidate routes prior to departure and during the trip.
- Recommend charging stops dynamically based on real-time station availability and compatibility.
- Incorporate weather and terrain data into route and energy calculations.
- Provide real-time route updates in response to changing traffic, weather, or charging conditions.

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise	Frequency of Use
EV Drivers	Individual electric vehicle owners planning personal trips who need reliable, energy-aware navigation.	Low to Moderate	Daily to Weekly
Fleet Operators	Organizations managing multiple electric vehicles who require route planning across a fleet to optimize energy costs and delivery schedules.	Moderate	Daily
Charging Station Providers	Operators of charging infrastructure who supply station location, availability, and pricing data to the platform via integration.	Moderate to High	Continuous (via API)
System Administrators	Technical staff responsible for platform configuration, monitoring, and maintenance.	High	As Needed

2.4 Operating Environment

The platform will be delivered as a responsive web application accessible through modern desktop and mobile browsers, complemented by a companion mobile application for use inside the vehicle. The backend will be implemented using a scalable cloud-hosted architecture capable of processing real-time data streams from multiple external APIs concurrently. The system's AI and optimization components will run on cloud infrastructure sized to support the computational demands of route scoring and energy prediction, with horizontal scaling available during periods of peak demand. All client applications will require an active internet connection to access live traffic, weather, and charging data.

2.5 Design and Implementation Constraints

- Third-party mapping, weather, and charging network APIs impose rate limits that constrain the frequency of route recalculation.
- Route optimization is fundamentally dependent on the availability and freshness of real-time data; degraded data feeds will reduce routing accuracy.
- The system must comply with applicable data privacy regulations when handling location data and vehicle telematics.
- Access to live vehicle battery state of charge depends on onboard GPS and sensor hardware, which may vary in accuracy across vehicle makes and models.

2.6 User Documentation

- A user manual covering account setup, trip planning, and charging stop navigation.
- In-app contextual help and guided tours for first-time users.
- API documentation for developers integrating third-party charging networks or fleet management systems.

2.7 Assumptions and Dependencies

- The platform assumes continued availability of third-party weather, traffic, and charging network APIs at acceptable service levels.
- The platform assumes users have a stable internet connection while planning and executing trips.
- The platform depends on accurate vehicle-specific energy consumption profiles being available or derivable for supported EV models.
- The platform assumes charging network partners provide sufficiently up-to-date station status data to support dynamic recommendations.

3. External Interface Requirements

3.1 User Interfaces

The platform will provide a route input screen where drivers enter their origin, destination, and optional preferences such as maximum detour time for charging. A live map view will display the recommended route, current vehicle position, and battery state of charge in real time. A charging suggestions panel will present recommended charging stops along the route, including estimated arrival state of charge, charging duration, and connector compatibility. All screens will be designed for quick readability while driving, with large touch targets and minimal required interaction.

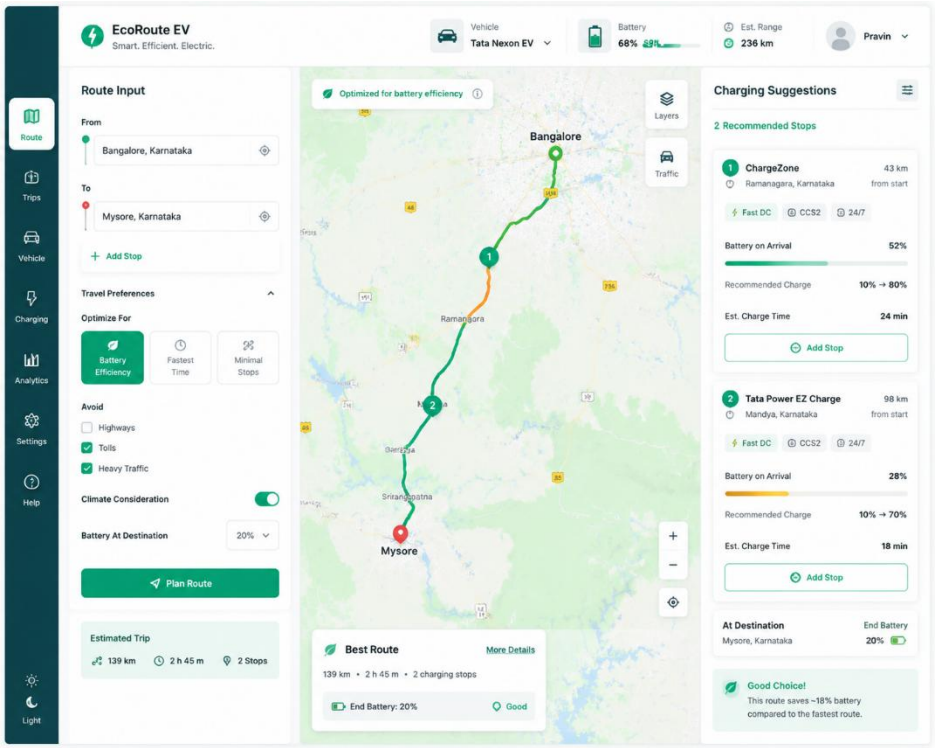


Figure 3. A representative screen layout showing the route input form, live map view, and charging suggestions panel.

3.2 Hardware Interfaces

The platform requires access to the vehicle's onboard GPS module to determine current location and track progress along the recommended route. Where supported, the platform will also

interface with the vehicle's battery management sensor system to obtain real-time state-of-charge data, enabling more accurate energy predictions than would be possible using estimated consumption alone. In cases where direct vehicle telematics access is unavailable, the platform will fall back to driver-supplied state-of-charge input.

3.3 Software Interfaces

Interface Name	Purpose	Data Exchanged	Protocol
Maps API	Base road network, distance, and turn-by-turn navigation data.	Route geometry, road attributes, elevation profile	REST / HTTPS
Weather API	Live and forecast weather conditions along the route.	Temperature, precipitation, wind speed and direction	REST / HTTPS
Charging Station API	Location, availability, and connector type of nearby charging stations.	Station location, status, connector type, pricing	REST / HTTPS
Vehicle Telematics API	Real-time vehicle battery state of charge and energy consumption data.	State of charge, battery health, energy usage history	REST / HTTPS or vehicle-specific SDK

3.4 Communications Interfaces

All communication between the platform's client applications and backend services will occur over HTTPS to ensure data is encrypted in transit. The system will expose a RESTful API for integration with external partners, using JSON as the primary data interchange format. Route and energy data will be synchronized with the vehicle at a frequency sufficient to keep displayed estimates current, targeting an update interval of no more than thirty seconds during active navigation. Authentication between the platform and third-party services will use industry-standard token-based mechanisms to ensure secure, authorized data exchange.

4. Functional requirements

4.1 Battery-Optimized Route Generation

Description and Priority

This feature calculates one or more candidate driving routes between a driver's specified origin and destination, evaluating each candidate on projected battery consumption rather than distance or time alone. The engine weighs traffic, terrain, and weather data to select the route that best minimizes total energy use while remaining within an acceptable travel-time bound.

Priority: High — this is the core function of the platform and all other features depend on it.

Stimulus/Response Sequences

1. The driver enters an origin and destination into the route input screen.
2. The system retrieves live traffic, terrain, and weather data for the relevant road network.
3. The optimization engine scores multiple candidate routes based on projected energy consumption.
4. The system displays the recommended route to the driver, along with estimated energy usage and arrival state of charge.
5. The driver confirms the route, and turn-by-turn navigation begins.

Functional Requirements

ID	Requirement
REQ-1.1	The system shall generate at least one candidate route between any valid origin and destination pair within the supported road network.
REQ-1.2	The system shall score each candidate route according to a weighted energy-consumption model that incorporates traffic, terrain, and weather inputs.
REQ-1.3	The system shall present the recommended route to the driver along with the projected energy consumption and estimated arrival state of charge.
REQ-1.4	The system shall allow the driver to view alternative candidate routes and their associated energy trade-offs.
REQ-1.5	The system shall complete route generation within ten seconds under normal network and data-availability conditions.

4.2 Energy Consumption Prediction

Description and Priority

This feature estimates the amount of battery charge that will be consumed over a given route segment, using a predictive model informed by vehicle-specific energy characteristics, road grade, driving speed, and environmental conditions. Accurate prediction allows the platform to determine whether a route is achievable on the vehicle's current charge and to plan charging stops accordingly.

Priority: High — accurate prediction underpins the reliability of every route and charging recommendation the platform makes.

Stimulus/Response Sequences

1. The system receives a candidate route and the vehicle's current battery state of charge.
2. The prediction model calculates expected energy consumption for each segment of the route.
3. The system aggregates segment-level predictions into a total energy estimate for the trip.

4. The system compares the total estimate against the vehicle's available charge to determine feasibility.
5. If the route is not feasible without recharging, the system flags the point at which charging becomes necessary.

Functional Requirements

ID	Requirement
REQ-2.1	The system shall predict energy consumption for each segment of a candidate route based on vehicle characteristics, terrain, and driving conditions.
REQ-2.2	The system shall aggregate segment-level energy predictions into a total trip-level energy estimate.
REQ-2.3	The system shall compare predicted energy consumption against the vehicle's current state of charge to determine route feasibility.
REQ-2.4	The system shall update energy predictions when underlying traffic, weather, or terrain data changes materially during the trip.
REQ-2.5	The system shall maintain a prediction accuracy such that actual consumption deviates from the estimate by no more than a configurable threshold under normal conditions.

4.3 Dynamic Charging Stop Recommendation

Description and Priority

This feature identifies and recommends charging stations along a route whenever the predicted energy consumption exceeds the vehicle's available charge. Recommendations account for station availability, connector compatibility, and the minimal detour required, ensuring that suggested stops integrate smoothly into the overall trip.

Priority: High — without reliable charging recommendations, drivers cannot complete routes that exceed their vehicle's single-charge range.

Stimulus/Response Sequences

1. The system determines that a route requires at least one charging stop based on the energy prediction.
2. The system queries the charging station API for candidates near the relevant point on the route.
3. The system filters candidates by connector compatibility and current availability.
4. The system ranks remaining candidates by minimal added detour time and presents the top recommendation to the driver.
5. The driver accepts the recommended stop, and the route is updated to include it.

Functional Requirements

ID	Requirement
REQ-3.1	The system shall identify when a planned route requires one or more charging stops based on predicted energy consumption.
REQ-3.2	The system shall query available charging stations near the relevant point on the route and filter results by connector compatibility.
REQ-3.3	The system shall rank candidate charging stations by minimal added detour time and current availability.
REQ-3.4	The system shall allow the driver to accept, reject, or manually select an alternative charging stop.
REQ-3.5	The system shall update the active route to include the selected charging stop and recalculate subsequent energy predictions accordingly.

4.4 Weather and Terrain Adaptation

Description and Priority

This feature incorporates live weather data and road elevation profiles into the route scoring and energy prediction models, since both factors materially affect real-world battery consumption.

Cold temperatures, precipitation, headwinds, and steep grades all increase energy draw beyond what distance-based estimates alone would suggest.

Priority: Medium — this feature significantly improves prediction accuracy but the platform can operate with reduced precision in its absence.

Stimulus/Response Sequences

1. The system retrieves current and forecast weather data for the geographic area covering the planned route.
2. The system retrieves elevation data for each segment of the candidate routes.
3. The energy prediction model adjusts consumption estimates based on temperature, precipitation, wind, and grade.
4. The system re-ranks candidate routes if weather or terrain conditions materially change the relative energy efficiency of the options.
5. The system displays a summary of the weather and terrain factors influencing the current recommendation.

Functional Requirements

ID	Requirement
REQ-4.1	The system shall retrieve current and forecast weather data for the geographic area covering a planned route.
REQ-4.2	The system shall retrieve elevation profile data for each segment of a candidate route.
REQ-4.3	The system shall adjust energy consumption predictions based on temperature, precipitation, wind conditions, and road grade.
REQ-4.4	The system shall re-rank candidate routes when weather or terrain data changes the relative energy efficiency of available options.
REQ-4.5	The system shall display a summary of the weather and terrain factors contributing to the current route recommendation.

4.5 Real-Time Route Updates

Description and Priority

This feature monitors conditions during an active trip and recalculates the route or charging plan when traffic incidents, weather changes, or charging station status updates materially affect the optimal path. The goal is to ensure the driver is always following the most energy-efficient route achievable given current conditions.

Priority: Medium — this feature enhances trip reliability but the platform remains functional with periodic rather than continuous updates.

Stimulus/Response Sequences

1. The system continuously monitors traffic, weather, and charging station data during an active trip.
2. The system detects a material change in one or more of these data sources.
3. The system recalculates the remaining route and energy prediction based on the updated data.
4. If the recalculated route differs meaningfully from the current route, the system notifies the driver of the proposed change.
5. The driver accepts or declines the updated route, and navigation continues accordingly.

Functional Requirements

ID	Requirement
REQ-5.1	The system shall monitor traffic, weather, and charging station data continuously during an active trip.
REQ-5.2	The system shall recalculate the remaining route and energy prediction when a material change in monitored data is detected.
REQ-5.3	The system shall notify the driver when a recalculated route offers a meaningfully different energy or time outcome than the current route.

ID	Requirement
REQ-5.4	The system shall allow the driver to accept or decline a proposed route update without interrupting active navigation.
REQ-5.5	The system shall limit the frequency of route recalculation to avoid excessive notifications, subject to a configurable minimum interval.

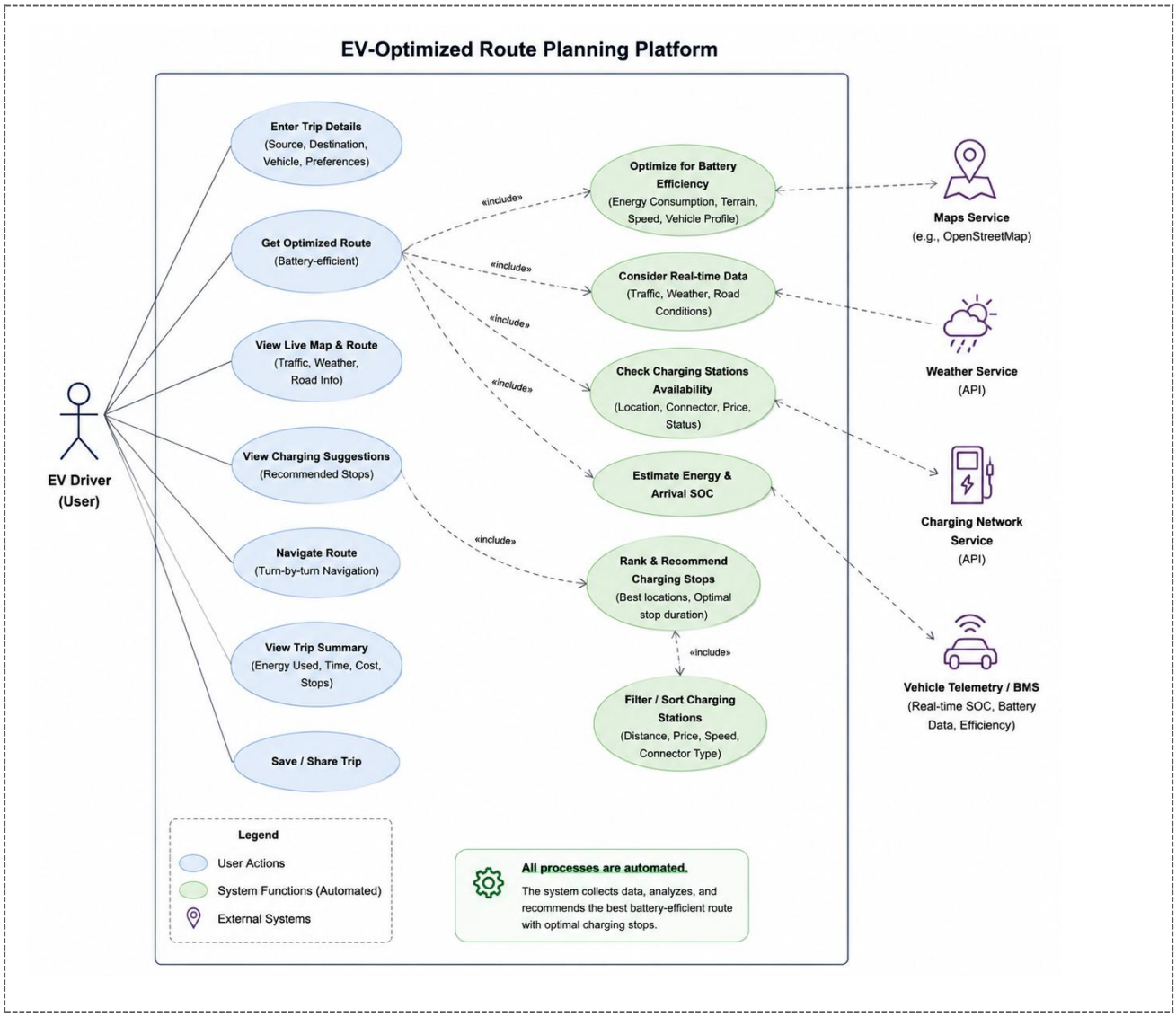


Figure 5. EV Driver, Fleet Operator, and Admin interactions with the system.

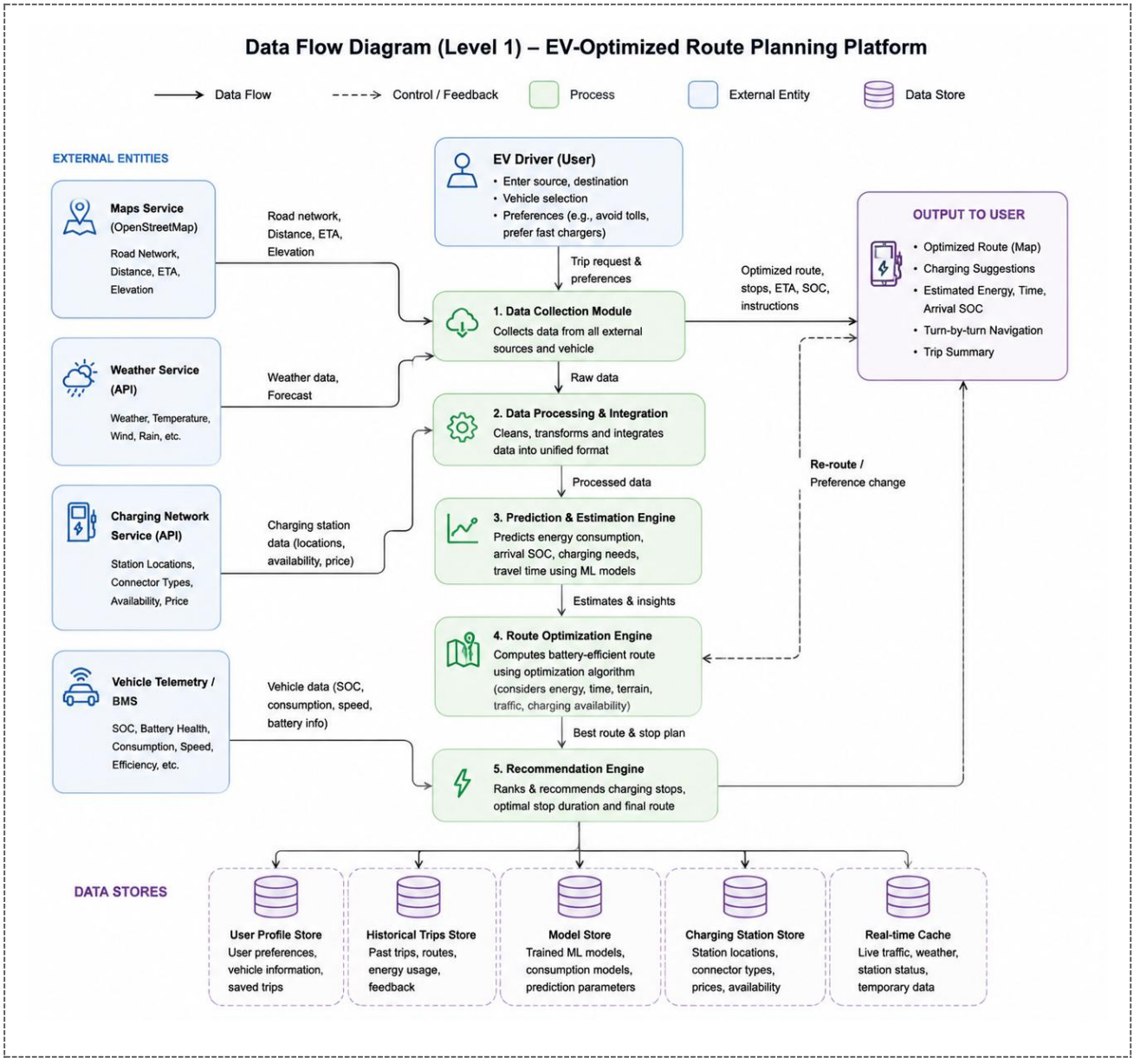


Figure 4. Input data sources, the route optimization engine, and output to the user.

5. Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement
NFR-1.1	The system shall generate an initial route recommendation within ten seconds under normal data-availability conditions.
NFR-1.2	The system shall recalculate an active route within five seconds of detecting a material change in traffic, weather, or charging data.
NFR-1.3	The system shall support at least 10,000 concurrent active navigation sessions without degradation in response time.

5.2 Safety Requirements

If charging station data is temporarily unavailable from an external provider, the system shall fall back to the most recently cached station data and clearly indicate to the driver that the information may not reflect real-time availability. In cases where no route can be generated that keeps the vehicle within a safe energy margin, the system shall alert the driver before departure rather than allowing navigation to begin on an infeasible route.

5.3 Security Requirements

All data transmitted between the client applications, backend services, and third-party APIs shall be encrypted using industry-standard transport encryption. User accounts shall be protected through secure authentication mechanisms, and API keys used to access third-party services shall be stored securely and never exposed to client applications. Access to vehicle telematics data shall be limited to the minimum scope necessary to support route and energy prediction functionality.

5.4 Software Quality Attributes

- **Scalability:** The platform shall be designed to scale horizontally to accommodate growth in user base and trip volume without requiring architectural redesign.

- **Reliability:** The platform shall maintain consistent route recommendation accuracy even when individual external data sources experience temporary degradation.
- **Usability:** The platform's interface shall be simple enough for drivers to interact with safely and quickly while minimizing distraction during active driving.
- **Maintainability:** The platform's routing and prediction components shall be modular, allowing individual data integrations to be updated or replaced without affecting unrelated system functionality.

5.5 Business Rules

- Only registered users shall be able to save route history and vehicle profiles for future trips.
- Charging stop recommendations shall only include stations verified as compatible with the driver's registered vehicle connector type.
- Route recalculation shall not be triggered more frequently than the configured minimum interval, to avoid excessive disruption to the driver.

6. Other Requirements

The platform shall maintain a database of route history, user vehicle profiles, and charging station records sufficient to support personalized recommendations and historical trip review. All handling of personal and location data shall comply with applicable data privacy and protection regulations in the jurisdictions where the platform operates. The system shall support internationalization of its user interface to accommodate future expansion into additional language markets, though initial release will target a single primary language.

Appendix A: Glossary

Term	Definition
EV	Electric Vehicle; a vehicle powered wholly or partly by an electric battery rather than an internal combustion engine.
Range Anxiety	The fear or concern that an electric vehicle's battery will be depleted before reaching a destination or suitable charging point.
API	Application Programming Interface; a defined set of protocols used for building and integrating application software.
SRS	Software Requirements Specification; a document that describes the intended functionality and constraints of a software system.
Telematics	The integrated use of telecommunications and informatics for the transmission and monitoring of data from a vehicle.
Route Optimization	The process of selecting the most efficient path between two points based on defined criteria such as energy consumption, time, or distance.
State of Charge (SoC)	The current level of charge in a battery, typically expressed as a percentage of full capacity.
Connector Compatibility	The match between a vehicle's charging port type and the connector type offered by a charging station.

Appendix B: Analysis Models

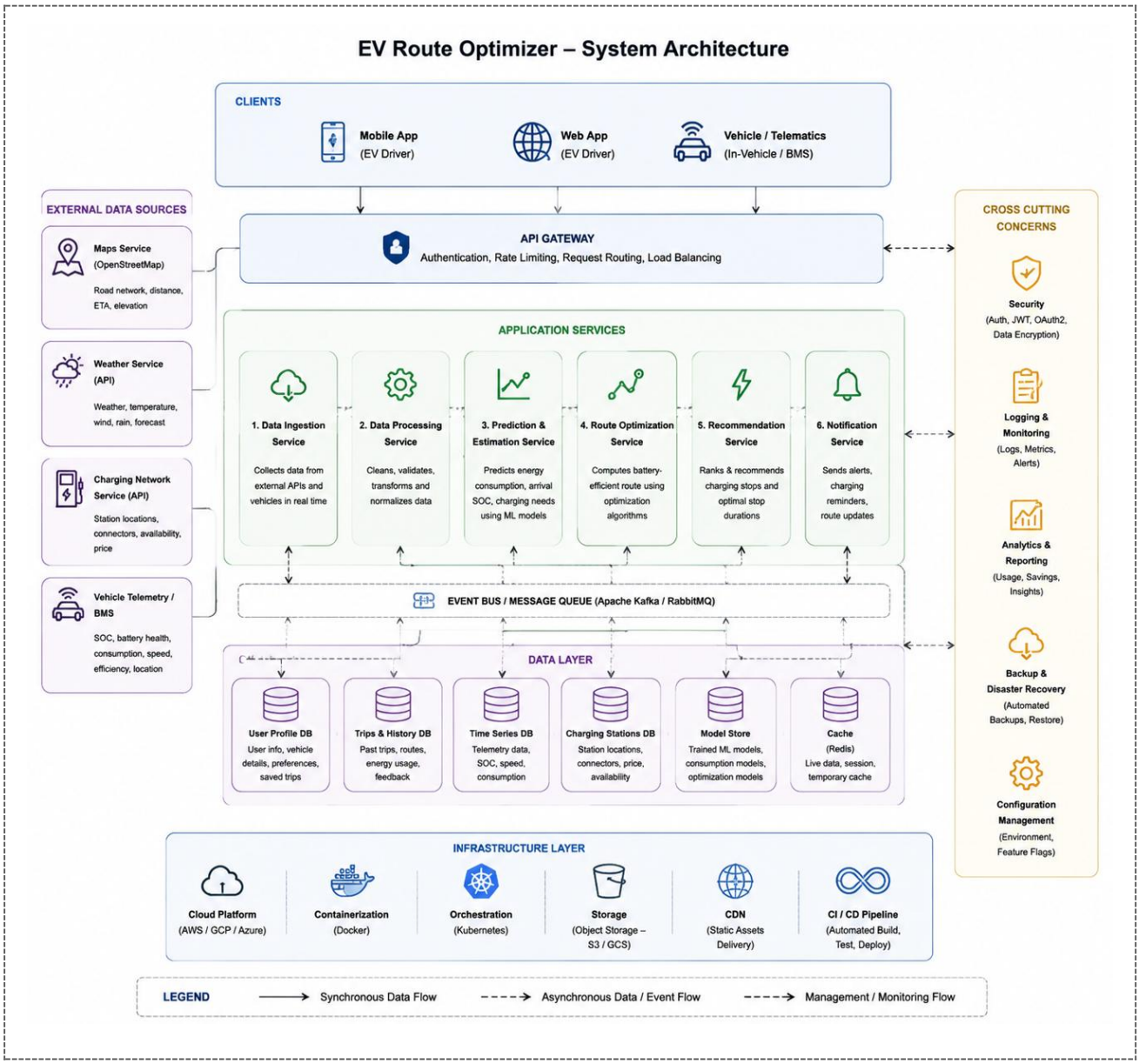


Figure 6. The overall software architecture, including client applications, backend services, and external integrations.

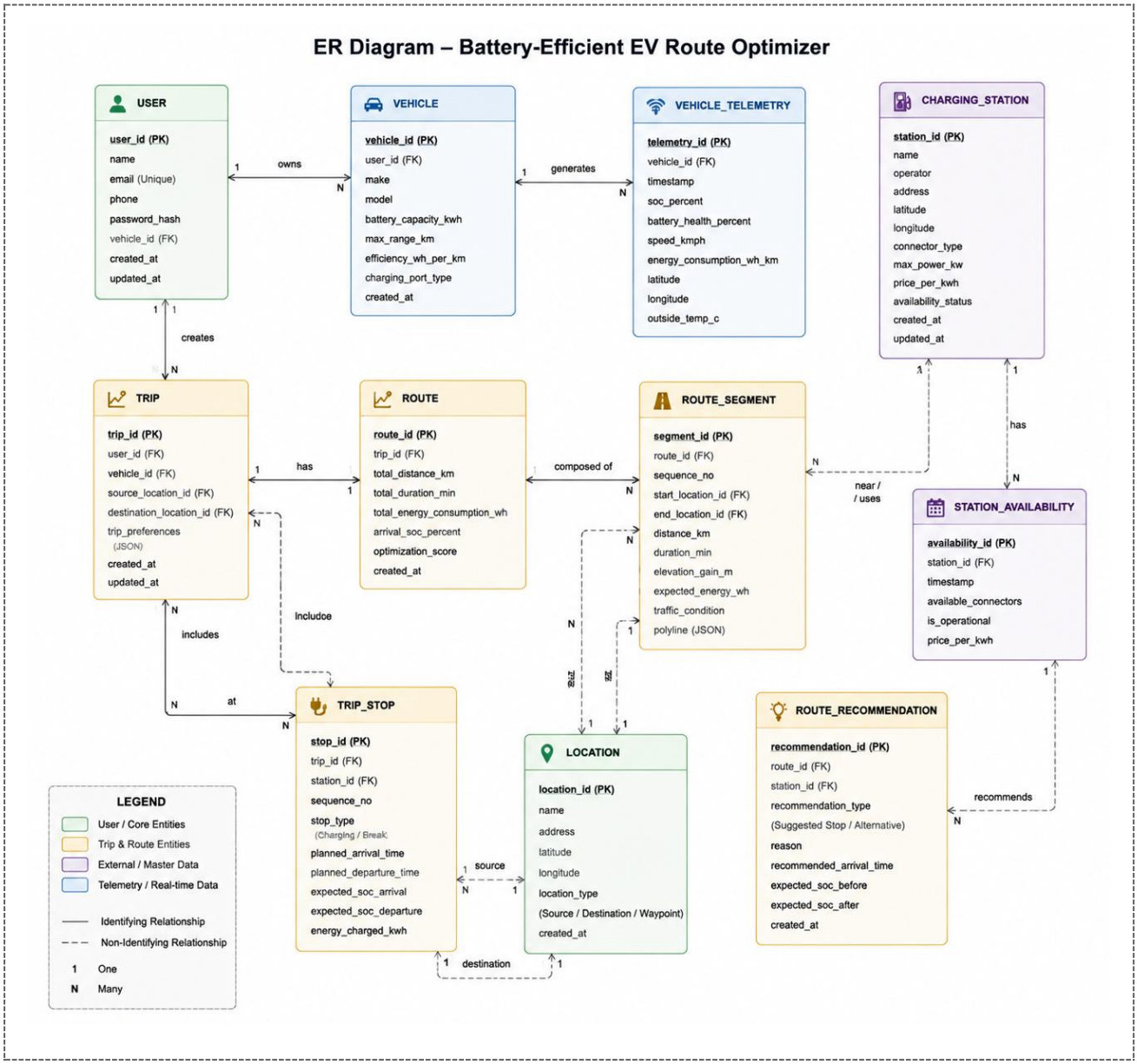


Figure 7. Entities: User, Vehicle, Charging_Station, Trip, Route,Location, Legend

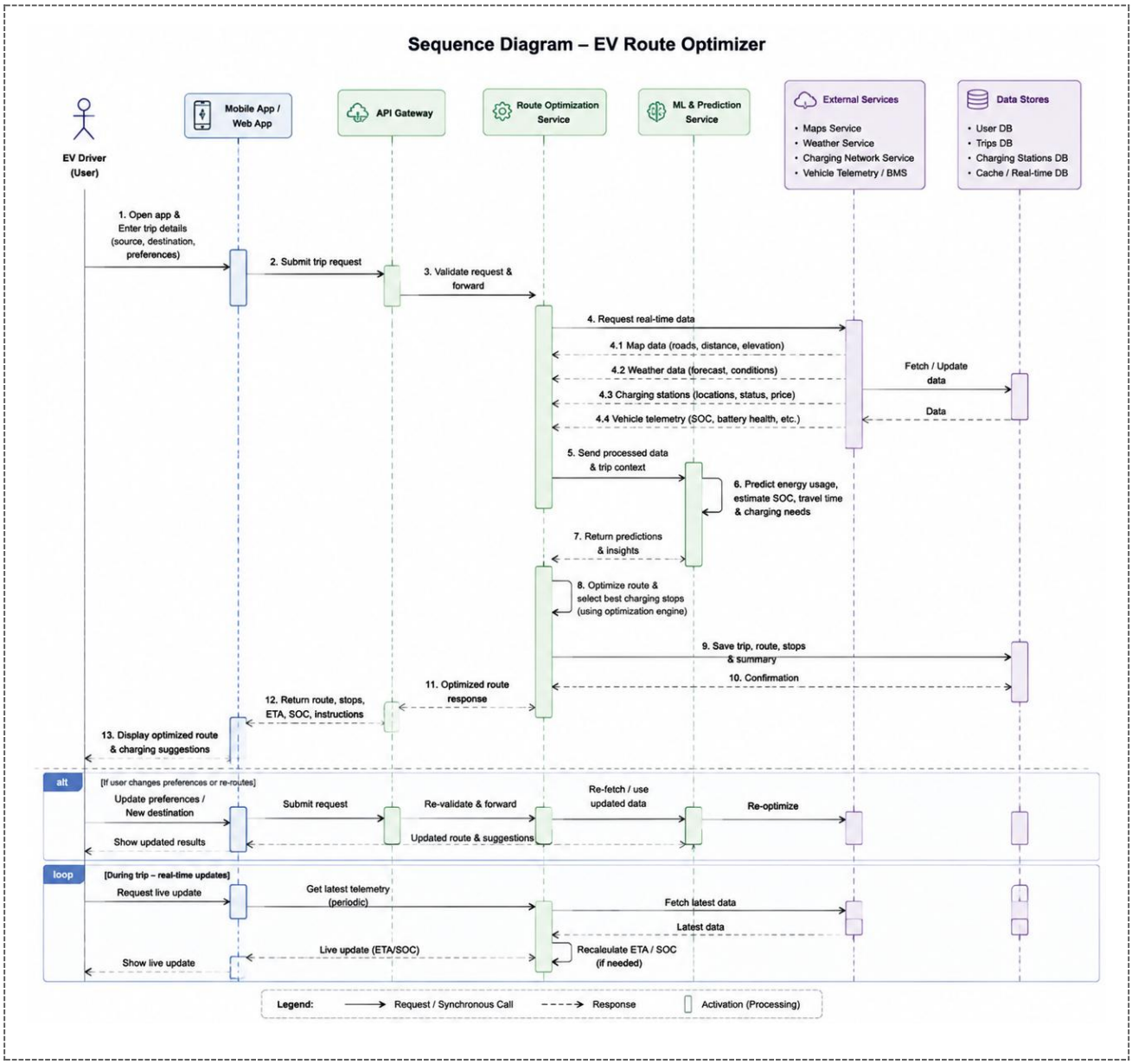


Figure 8. Sequence Diagram.

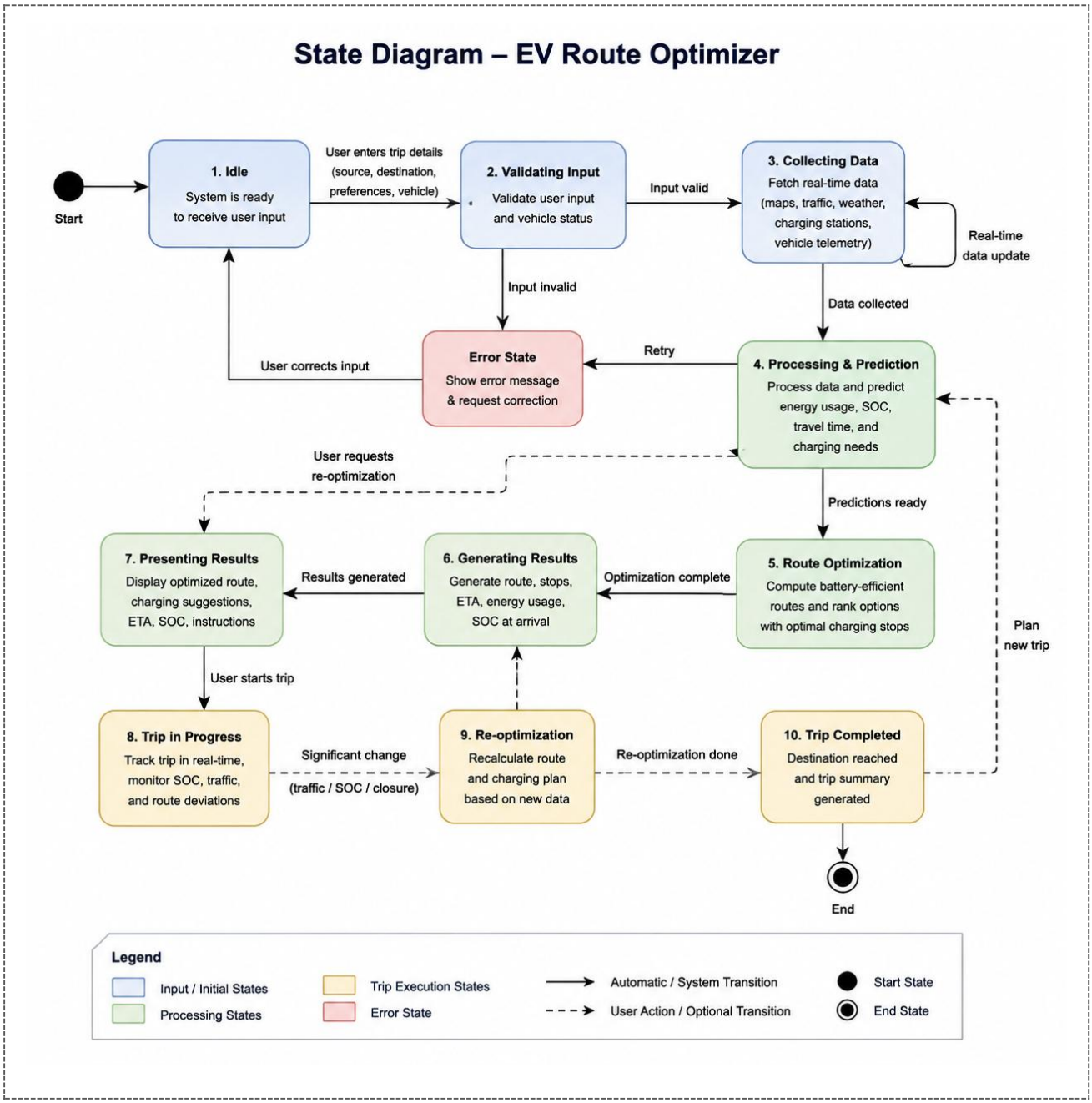


Figure 9. State Diagram - EV Route diagram

Appendix C: To Be Determined (TBD) List

TBD ID	Description	Section Reference	Status
TBD-1	Final selection of primary mapping data provider.	Section 3.3	Open
TBD-2	Confirmed list of supported vehicle makes and models at launch.	Section 4.2	Open
TBD-3	Target market and initial supported language(s).	Section 6	Open
TBD-4	Specific data retention period for trip and location history.	Section 6	Open